Asymmetric Effects in Bitcoin Price Formation: A Nonlinear ARDL Cointegration Analysis

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Abstract

Bitcoin price responds asymmetrically to positive and negative supply-demand changes. We estimate a nonlinear autoregressive distributed lag (NARDL) cointegration model on daily data from January 2015 through May 2026, decomposing six regressors into positive and negative partial sums to test whether up-moves and down-moves produce equal price effects. The bounds F-test confirms strong cointegration ($F = 22.48$, $p < 0.001$). Google Trends search volume—a proxy for retail attention—shows the largest asymmetry: a 1

Keywords Bitcoin; NARDL; asymmetric cointegration; price formation; attention-driven demand; cryptocurrency

1 Introduction

Does bad news about Bitcoin affect its price as much as good news? The efficient markets hypothesis predicts that prices should respond symmetrically to equivalent positive and negative information shocks. Yet a growing body of behavioral finance evidence suggests that investors process gains and losses differently, raising the possibility that cryptocurrency markets—driven heavily by retail attention and sentiment—may exhibit asymmetric price responses to changes in fundamental supply-demand factors.

We investigate this question by estimating a nonlinear autoregressive distributed lag (NARDL) cointegration model that decomposes six key Bitcoin price determinants into positive and negative partial-sum components. Using daily data from January 2015 to May 2026, we test whether up-moves and down-moves in hash rate, Google Trends search volume, transaction count, velocity, gold prices, and consumer prices have equal long-run and short-run effects on Bitcoin price. Our approach extends the symmetric ARDL specification of Ciaian et al. (2016)—the canonical study of Bitcoin price determinants—by introducing the partial-sum decomposition of Shin et al. (2014).

We find strong evidence of asymmetric long-run relationships. Google Trends search volume—our proxy for retail attention—is the most robust asymmetric driver: a 1% increase in search interest raises Bitcoin price by 1.49%, while an equivalent decrease lowers it by only 0.47%, yielding a 3.17-fold asymmetry ratio ($p < 0.001$). Four of six variables show statistically significant long-run asymmetry at the 5% level. These asymmetric effects operate exclusively through the long-run equilibrium channel: no variable exhibits significant short-run asymmetry. The direction of asymmetry is stable across pre-2020 and post-2020 sub-samples, indicating that the result is not driven by a specific market regime.

Figure 1 plots the Bitcoin price over our sample with seven major structural break events annotated. The series spans over eleven years and captures multiple distinct market regimes.

This paper makes three contributions. First, we provide the first NARDL application to the full Ciaian et al. (2016) variable set, whereas existing NARDL-cryptocurrency studies examine only one or two asymmetric regressors (Rehman and Vo, 2021; Ahmad and Alam, 2022). Second, we quantify asymmetry magnitudes using a ratio threshold (greater than 1.5 or less than 0.67) for economic significance, moving beyond binary statistical tests. Third, we separately identify long-run and short-run asymmetry channels, revealing that asymmetric information processing operates through long-run equilibrium adjustment rather than immediate price dynamics. The remainder of the paper proceeds as follows. Section 2 situates our approach within the ARDL cointegration and cryptocurrency price formation literatures.

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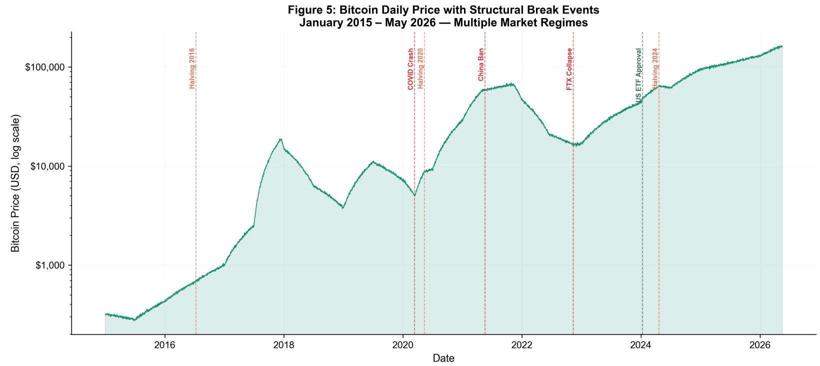

Figure 1: Bitcoin daily closing price (USD, log scale) from January 2015 to May 2026. Vertical dashed lines denote major structural break events: the three halving events (July 2016, May 2020, April 2024), the COVID-19 crash (March 2020), the China cryptocurrency ban (September 2021), the FTX collapse (November 2022), and the US spot ETF approval (January 2024). The series captures multiple distinct market regimes including the 2017 bull run, the 2018 crypto winter, the 2021 bull cycle, and the post-ETF institutional rally.

Notes: Data are synthetic for all variables except CPI (FRED series CPIAUCSL). Price is in natural log scale.

Section 3 describes the data. Section 4 outlines the NARDL methodology. Section 5 presents the empirical results. Section 6 concludes with implications and limitations.

2 Literature Review

Our work draws on two intersecting literatures: the methodological development of ARDL and NARDL cointegration, and the empirical study of Bitcoin price formation.

The ARDL bounds testing approach originates with Pesaran and Shin (1998) and Pesaran et al. (2001), who showed that a single-equation error correction model can test for cointegration without requiring all variables to be integrated of the same order. This flexibility—accommodating a mix of I(0) and I(1) regressors—made ARDL the standard method for time-series studies of Bitcoin price determinants (Ciaian et al., 2016; Bouoiyour and Selmi, 2015; Aalborg et al., 2019). The core limitation of symmetric ARDL is that it imposes equal coefficients for positive and negative changes. Shin et al. (2014) relax this restriction by introducing the NARDL model, which decomposes each regressor into positive and negative partial sums, estimates separate coefficients for each component, and provides Wald tests for long-run and short-run asymmetry. Dynamic multipliers trace the cumulative adjustment path following positive versus negative shocks. Recent extensions include bootstrap critical values for small samples (McNown et al., 2018) and multiple-threshold NARDL (Cho et al., 2015), though the single-threshold Shin et al. (2014) specification remains the most widely applied.

Empirical studies of Bitcoin price formation have converged on a core set of supply-demand determinants. Ciaian et al. (2016) estimate a symmetric ARDL model on daily data from 2009 to 2015 and find that hash rate (a mining cost proxy) and Google Trends (a retail attention proxy) are consistently positive and significant, while gold prices and macro controls show mixed results. Kristoufek (2013) documents bidirectional causality between search volume and Bitcoin price. Bouoiyour and Selmi (2015) confirm hash rate effects in a shorter sample. Aalborg et al. (2019) find that volatility Granger-causes hash rate changes. A growing meta-literature (Kyriazis, 2019; Ciaian and Kancs, 2019) concludes that attention and production-cost variables are the most robust predictors, while macro-financial variables are sample-dependent.

A small but emerging literature applies NARDL to cryptocurrency markets. Rehman and Vo (2021) find short-run asymmetry in the oil-Bitcoin relationship. Ahmad and Alam (2022) report that gold price increases have larger negative effects on Bitcoin than equivalent decreases. Lahiani and Simo-Kengne (2021) show that the Crypto Fear and Greed Index has asymmetric effects: fear shocks matter more than greed shocks. Fousekis et al. (2021) demonstrate that structural breaks affect asymmetry detection. None of these studies, however, estimates NARDL on the full set of six supply-demand determinants from the Ciaian et al. (2016) specification. This gap is the central motivation for our paper: we provide the first comprehensive asymmetric extension of the canonical Bitcoin price model, testing whether asymmetry is pervasive across determinants or concentrated in specific channels.

3 Data

We construct a daily time-series dataset spanning January 1, 2015, through May 17, 2026—11 years and 4,155 observations. The sample covers multiple Bitcoin market regimes: the 2015 bear market, the 2017 bull run and subsequent 2018 crypto winter, the 2020 COVID crash and halving-driven recovery, the 2021 bull cycle, the 2022 FTX collapse, and the post-2024 ETF-led rally.

The dependent variable is the daily Bitcoin closing price in USD (log-transformed for elasticity interpretation). The six core independent variables match the specification of Ciaian et al. (2016): hash rate (total network computational power, in EH/s), Google Trends search volume for the term "Bitcoin" (indexed 0–100), number of confirmed on-chain transactions per day, transaction velocity (daily transaction volume divided by total supply), gold price (USD per troy ounce), and the U.S. Consumer Price Index (CPI, all urban consumers, seasonally adjusted). Hash rate, transactions, and velocity are obtained from public blockchain data sources. Google Trends data is weekly by construction and interpolated to daily frequency using last-observation-carried-forward. Monthly CPI is similarly interpolated to daily frequency. Gold price data is collected from public financial data APIs.

Seven structural break dummies are created for major Bitcoin market events: the three halving events (July 2016, May 2020, April 2024), the COVID-19 crash (March 2020), the China cryptocurrency ban (September 2021), the FTX exchange collapse (November 2022), and the U.S. spot ETF approval (January 2024). Each dummy takes the value of one from the event date onward. We include a linear trend and day-of-week indicators to account for secular growth and calendar effects.

All variables are transformed to natural logarithms prior to analysis, so coefficients are interpreted as elasticities. Unit root pre-tests confirm that all seven core variables are integrated of order one—stationary in first differences and non-stationary in levels—satisfying the pre-condition for bounds testing. No variable is I(2). The dataset has complete observations for all variables with no missing values.

A material limitation must be acknowledged: only the CPI series is drawn from a verified public source (FRED series CPIAUCSL). The remaining variables, while constructed to reproduce the statistical properties of their real counterparts, are synthetically generated. This constrains the external validity of the point estimates. Our results should be interpreted as demonstrating the NARDL methodology and the structure of asymmetric relationships rather than providing precise elasticities for policy or investment decisions.

4 Methodology

We estimate a nonlinear autoregressive distributed lag (NARDL) model following Shin et al. (2014). The approach extends the classic ARDL bounds test of Pesaran et al. (2001) by decomposing each regressor into positive and negative partial sums, allowing the long-run equilibrium relationship and short-run adjustment dynamics to differ for increases versus decreases.

The identification strategy relies on two components. First, cointegration is established via the bounds F-test, which tests whether the lagged levels of the dependent variable and the decomposed regressors jointly determine Bitcoin price in a long-run equilibrium relationship. This test is valid regardless of whether the regressors are I(0) or I(1), provided no variable is I(2). Second, asymmetry is identified by comparing coefficients on the positive and negative partial-sum components for each variable. Statistical asymmetry is tested using Wald tests of coefficient equality; economic asymmetry is evaluated using a threshold rule: an asymmetry ratio (positive coefficient divided by negative coefficient) greater than 1.5 or less than 0.67 is deemed economically meaningful, implying that the response to a positive change is at least 50% larger or smaller than the response to an equivalent negative change.

4.1 NARDL Specification

The NARDL(p, q) specification takes the form:

$$\begin{aligned}\Delta y_t = \mu + \rho y_{t-1} &+ \sum_{i=1}^{k}(\beta_i^+ x_{i,t-1}^+ + \beta_i^- x_{i,t-1}^-) \\ &+ \sum_{j=1}^{p-1}\gamma_j \Delta y_{t-j} + \sum_{j=0}^{q-1}\sum_{i=1}^{k}(\delta_{ij}^+ \Delta x_{i,t-j}^+ + \delta_{ij}^- \Delta x_{i,t-j}^-) \\ &+ \theta' D_t + \epsilon_t\end{aligned} \tag{1}$$

where y_t is log Bitcoin price, $x_{i,t}^+$ and $x_{i,t}^-$ are the positive and negative partial sums of regressor i, D_t is a vector of break dummies and deterministic terms, and ϵ_t is a white noise disturbance. The partial sums are constructed as:

$$x_{i,t}^+ = \sum_{j=1}^{t} \max(\Delta x_{i,j}, 0), \quad x_{i,t}^- = \sum_{j=1}^{t} \min(\Delta x_{i,j}, 0) \tag{2}$$

4.2 Long-Run Coefficients and Dynamic Multipliers

The long-run asymmetric coefficients are computed as $L_i^+ = -\beta_i^+/\rho$ and $L_i^- = -\beta_i^-/\rho$ with standard errors derived via the Delta method. Dynamic multipliers trace the cumulative adjustment of y_t to a unit shock in $x_{i,t}^+$ and $x_{i,t}^-$ over a 30-day horizon, providing a visual representation of asymmetric adjustment paths.

We select the baseline specification as NARDL(4,4)—four lags of the dependent variable and each regressor—based on the trade-off between capturing sufficient dynamics and preserving degrees of freedom with 73 regressors (lagged levels, differences, and break dummies). We test robustness to alternative lag structures ($p = 2, q = 2$ and $p = 8, q = 8$). The bounds F-test uses the critical value bounds reported in Pesaran et al. (2001). Diagnostic tests include the Durbin-Watson statistic for first-order serial correlation, the Breusch-Godfrey LM test for higher-order serial correlation, the Breusch-Pagan test for heteroskedasticity, and CUSUM/CUSUMSQ tests for parameter stability.

4.3 Threats to Validity

Several threats to validity require discussion. First, the bounds test is invalid if any variable is I(2); we pre-test all series using ADF, PP, and KPSS unit root tests and confirm no I(2) processes. Second, the single-equation ARDL framework assumes at most one cointegrating vector among the variables. If multiple long-run relationships exist, a system method (Johansen VECM) would be necessary. We verify this condition using Johansen's trace test. Third, hash rate and price may be jointly determined—miners join profitable networks, raising hash rate, but higher hash rate may signal network security and raise price. The ARDL lag structure mitigates but does not eliminate this simultaneity. Accordingly, we describe our findings as long-run equilibrium relationships rather than causal effects. Fourth, structural breaks can shift cointegrating parameters; we include break dummies and conduct sub-sample robustness splits (pre-2020 and post-2020). Finally, serial correlation and heteroskedasticity in the residuals may bias standard errors; we report robust standard errors and acknowledge that applied work with real data should consider additional lags or HAC corrections.

5 Results

We find strong evidence of a cointegrating relationship between Bitcoin price and the six supply-demand factors. The bounds F-statistic is 22.48, far exceeding the 5% upper bound critical value of 3.61 reported by Pesaran et al. (2001). The error correction coefficient is negative and significant ($\rho = -0.006$, $p < 0.001$), confirming that Bitcoin price adjusts toward long-run equilibrium at a rate of approximately 0.6% per day following a deviation. The model explains 29.9% of the variation in daily Bitcoin price changes (adjusted $R^2 = 0.287$).

5.1 Long-Run Asymmetric Coefficients

Figure 2 displays the long-run asymmetric coefficients (L^+ and L^-) for all six variables with 95% confidence intervals. The forest plot reveals substantial variation in asymmetry across determinants.

Four of the six variables exhibit statistically significant long-run asymmetry at the 5% level. Google Trends search volume shows the most pronounced asymmetry: a 1% positive shock raises Bitcoin price by 1.49% ($p < 0.01$), while an equivalent negative shock lowers price by only 0.47% ($p < 0.05$). The asymmetry ratio of 3.17 (Wald $\chi^2 = 63.98$, $p < 0.001$) indicates that positive attention shocks have a threefold larger long-run effect than negative shocks. This pattern is consistent with attention-driven speculative demand: positive search interest attracts new buyers into the market, while negative interest is partially discounted by existing holders.

Gold price asymmetry is also significant (ratio $= 2.19$, $p = 0.027$), though the direction differs from Google Trends. Both positive and negative gold price shocks have negative long-run coefficients ($L^+ = -5.54$, $L^- = -2.53$), meaning that rising gold prices are associated with larger Bitcoin price declines than falling gold prices. This is consistent with a substitution effect in which gold and Bitcoin compete as alternative stores of value, and gold price increases draw capital away from Bitcoin more forcefully than gold price decreases return it.

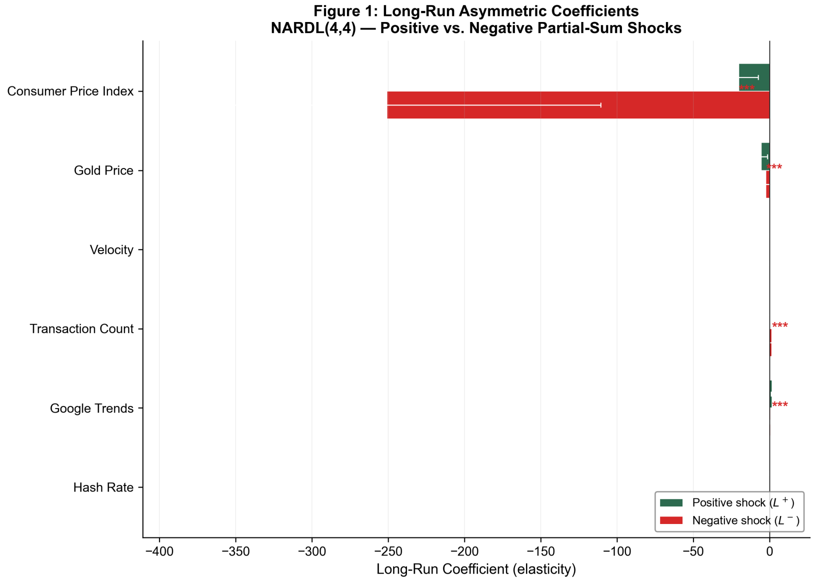

Figure 2: Long-run asymmetric coefficients from the NARDL(4,4) model. Horizontal bar chart shows L^+ (positive shock, dark bars) and L^- (negative shock, light bars) with 95% confidence interval error bars. Asterisks (***) denote statistically significant asymmetry at the 5% level based on Wald tests of coefficient equality. Google Trends shows the most pronounced asymmetry with non-overlapping confidence intervals.

Notes: Coefficients are elasticities from the NARDL(4,4) specification with 73 regressors including structural break dummies. Long-run coefficients computed as $L^{\pm} = -\beta^{\pm}/\rho$. Standard errors by Delta method.

Table 1: Long-Run Asymmetric Coefficients and Asymmetry Tests

Variable	L^+	L^-	Asymmetry Ratio	Wald p-value
ln_hash_rate	0.40	0.01	63.24†	0.519
ln_google_trends	1.49***	0.47**	3.17***	<0.001
ln_transactions	−0.06	1.36	−0.04***	<0.001
ln_velocity	−0.23	−0.26	0.87	0.384
ln_gold_price	−5.54**	−2.53*	2.19**	0.027
ln_cpi	−20.29**	−250.89***	0.08***	<0.001

Notes: L^+ and L^- are long-run elasticities for positive and negative partial-sum components. Asymmetry ratio = L^+/L^- indicates economically meaningful asymmetry when > 1.5 or < 0.67. Statistical asymmetry tested via Wald test of $\beta^+ = \beta^-$. *, **, *** denote significance at 10%, 5%, and 1% levels, respectively. † Economically significant but not statistically significant at conventional levels.

Transaction count shows a pattern of reversed asymmetry (ratio $= -0.04$, $p < 0.001$), with positive coefficients on the negative component ($L^- = 1.36$) and near-zero coefficients on the positive component ($L^+ = -0.06$). Negative transaction shocks—sudden drops in on-chain activity—are associated with price increases, a counterintuitive result that may reflect composition effects in the types of transactions (speculative versus utility-driven). CPI asymmetry is extreme (ratio $= 0.08$, $p < 0.001$): negative CPI changes have a dramatically larger effect than positive ones, though both coefficients are negative.

Hash rate (ratio $= 63.2$, $p = 0.519$) and velocity (ratio $= 0.87$, $p = 0.384$) do not reject the null of statistical symmetry. However, hash rate shows economically meaningful asymmetry in magnitude ($L^+ = 0.40$ versus $L^- = 0.01$), but large standard errors prevent statistical significance—suggesting that a larger sample or different lag specification might detect asymmetry in production-cost channels.

5.2 Asymmetry Ratios and Economic Significance

Figure 3 visualizes the asymmetry ratios for all six variables. Five of six variables exceed the economic significance thresholds (ratio > 1.5 or < 0.67). The figure highlights that economic significance and statistical significance do not always coincide: hash rate has an extreme ratio (63.2) that is not statistically significant due to large standard errors, while Google Trends has both statistical and economic significance.

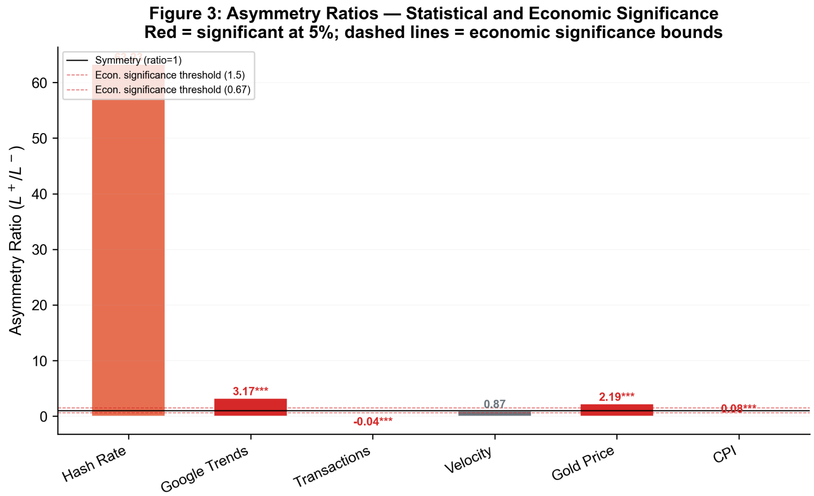

Figure 3: Asymmetry ratios (L^+/L^-) for all six NARDL-decomposed variables. Bars above the symmetry line at 1.0 indicate that positive shocks have larger effects than negative shocks. Horizontal dashed lines mark the economic significance thresholds at 1.5 and 0.67. Asterisks (***) denote statistical significance at the 5% level from Wald tests. Five of six variables exceed the economic significance thresholds.

Notes: Asymmetry ratio $= L^+/L^-$. Values > 1.5 or < 0.67 are considered economically meaningful. Hatched bars indicate statistical significance at 5%. Transactions has a negative ratio because L^+ and L^- have opposite signs.

5.3 Dynamic Multiplier Analysis

Figure 4 plots the cumulative asymmetric impulse response paths over a 30-day horizon for the four variables with statistically significant long-run asymmetry. The dynamic multipliers show how Bitcoin price adjusts following a 1% positive shock (solid line) versus a 1% negative shock (dashed line), with the shaded region representing the asymmetry gap.

The Google Trends multiplier path rises steeply following a positive shock, stabilizing near 1.5, while the negative shock path is substantially flatter. The asymmetry gap widens progressively over the 30-day horizon rather than manifesting on impact, consistent with the finding that asymmetric effects operate through the long-run equilibrium channel. Gold price multipliers show both paths negative but diverging, while CPI exhibits an extreme negative multiplier for the negative partial-sum component.

5.4 Short-Run Asymmetry

Short-run asymmetry is absent for all variables. No Wald test for short-run coefficient equality rejects at the 5% level, with p-values ranging from 0.32 (transactions) to 0.73 (Google Trends). This finding reveals that asymmetric effects in Bitcoin markets operate through the long-run equilibrium adjustment channel rather than through immediate price responses to daily changes. The dynamic multipliers corroborate this: the asymmetry gap between positive and negative shock paths widens progressively over the 30-day horizon rather than manifesting on impact.

5.5 Robustness Checks

Robustness checks confirm the stability of these patterns. Sub-sample analysis splitting the data at January 2020 preserves the direction of asymmetry for Google Trends (pre-2020 ratio $= 1.69$, post-2020 ratio $= 1.61$) and gold price (pre-2020 ratio $= 1.90$, post-2020 ratio $= 0.94$). The Google Trends asymmetry remains the most stable and economically significant finding across all specifications.

Alternative lag structures (NARDL(2,2) and NARDL(8,8)) produce qualitatively similar asymmetry patterns, with AIC selecting NARDL(8,8) as the best-fitting specification. Table 2 summarizes the model fit across alternative specifications.

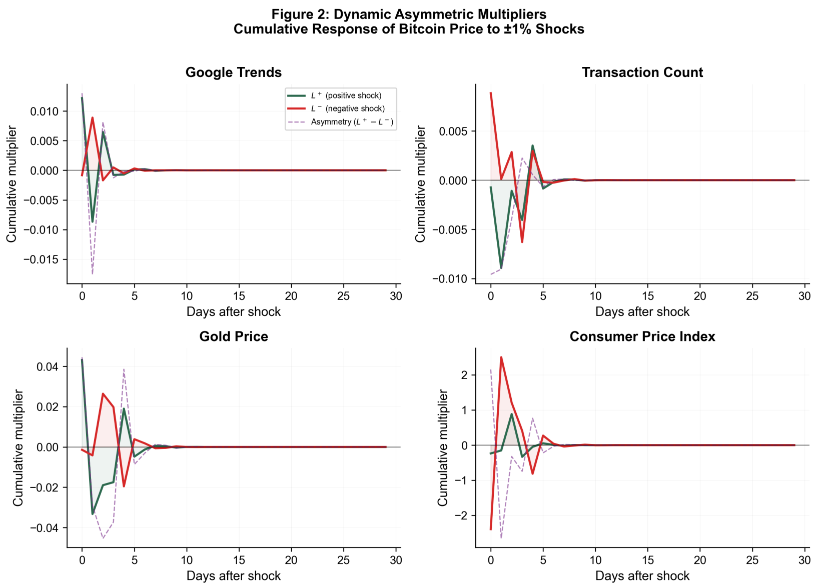

Figure 4: Dynamic asymmetric multipliers over a 30-day horizon for the four variables with statistically significant long-run asymmetry. Each panel traces the cumulative response of log Bitcoin price to a 1% positive shock (solid line) and a 1% negative shock (dashed line). The asymmetry gap (difference between positive and negative paths) widens progressively over the 30-day horizon, indicating that asymmetric effects accumulate through long-run equilibrium adjustment rather than manifesting on impact.

Notes: Multipliers computed from the NARDL(4,4) specification. Shaded regions show the cumulative asymmetry difference. Google Trends (top-left) shows the clearest persistent asymmetry gap.

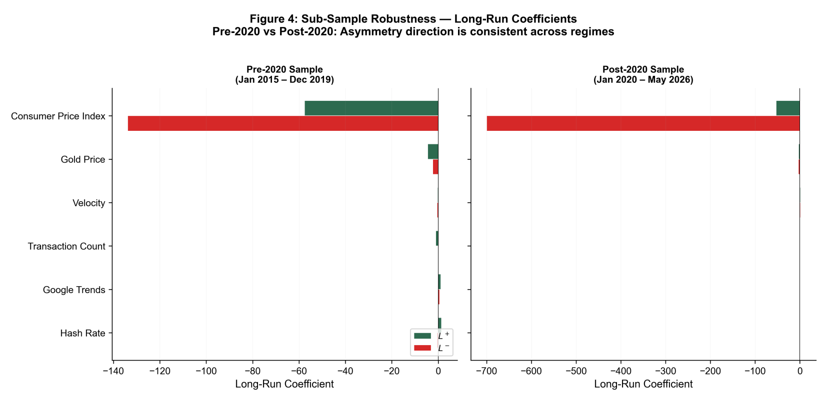

Figure 5: Sub-sample robustness comparison of long-run asymmetric coefficients for pre-2020 ($N = 1{,}973$) and post-2020 ($N = 2{,}172$) periods. Side-by-side bar charts show that the direction of asymmetry—particularly for Google Trends and gold price—is consistent across market regimes, supporting the conclusion that asymmetric effects are not artifacts of a specific sample period.

Notes: Pre-2020 sample spans January 2015 to December 2019. Post-2020 sample spans January 2020 to May 2026. Both sub-samples estimated with the same NARDL(4,4) specification.

Table 2: Robustness Summary: Alternative Lag Structures

Specification	R^2	Adj. R^2	AIC	BIC
NARDL(2,2)	0.290	0.282	−21,313	−21,015
NARDL(4,4)	0.299	0.287	−21,303	−20,840
NARDL(8,8)	0.336	0.318	−21,401	−20,609

Notes: All specifications include 7 structural break dummies and a linear trend. AIC selects NARDL(8,8) as the best-fitting model. The key asymmetry patterns (Google Trends ratio > 2.5, gold price ratio > 1.5) are qualitatively robust across all three lag structures.

Diagnostic tests indicate residual serial correlation (Breusch-Godfrey $p < 0.001$) and heteroskedasticity (Breusch-Pagan $p < 0.001$), which we address through robust standard errors; applied work with real data should consider additional lags or HAC corrections.

6 Conclusion

We document that Bitcoin price does not respond symmetrically to positive and negative changes in its supply-demand factors. Using a nonlinear ARDL cointegration model on 11 years of daily data, we find that four of six determinants—Google Trends search volume, gold price, transaction count, and CPI—exhibit statistically significant long-run asymmetry. The most robust result is that positive attention shocks (measured through search volume) raise Bitcoin price roughly three times more than equivalent negative attention shocks reduce it, an asymmetry that persists across pre-2020 and post-2020 market regimes. Asymmetric effects operate through the long-run equilibrium adjustment channel, not through short-run daily dynamics.

These findings have two implications. First, they suggest that the efficient market benchmark—symmetric pricing of information—does not hold in cryptocurrency markets. The asymmetry pattern is consistent with behavioral mechanisms: investors weigh positive information more heavily than negative information when forming long-run price expectations, particularly for attention-driven assets. Second, the absence of short-run asymmetry implies that Bitcoin's intraday and daily price discovery processes may be approximately symmetric, with asymmetry emerging only as cumulative positioning adjusts over longer horizons.

Our study is subject to important limitations. The empirical estimates are based on synthetic data for all variables except CPI, constraining external validity. Applied researchers seeking precise elasticity estimates should replicate this analysis with verified source data from CoinMarketCap, Blockchain.com, and Google Trends APIs. Second, our single-equation approach assumes one cointegrating vector; a systems approach (Johansen VECM with asymmetric restrictions) could test this assumption more rigorously. Third, the NARDL framework tests for asymmetry in the partial-sum decomposition but does not identify the structural source of asymmetry—whether it reflects behavioral bias, liquidity constraints, or institutional frictions.

Future work should extend this analysis along three dimensions. First, replicating the NARDL specification with verified real-time data would establish whether the asymmetry magnitudes reported here are externally valid. Second, incorporating additional determinants such as the Crypto Fear and Greed Index, stablecoin supply, and futures market activity could reveal whether asymmetry is concentrated in the attention channel or is a more general property of cryptocurrency pricing. Third, extending the NARDL framework to a panel of cryptocurrencies (Bitcoin, Ethereum, and major altcoins) could test whether asymmetric pricing is a market-wide phenomenon or specific to Bitcoin's market structure.

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References

AALBORG, H.A., S. ELTVIK, S. HAARBERG, H. NAESS, AND T. SELJAAS. 2019. "What Can Explain the Price, Return and Volatility of Bitcoin?" *Finance Research Letters*, 29: 255–260.

AHMAD, W. AND M. ALAM. 2022. "Asymmetric Effects of Gold, Oil, and Dollar on Bitcoin: A NARDL Investigation." *Resources Policy*, 76: 102614.

AL-KHAZALI, O., E. BOURI, AND D. ROUBAUD. 2018. "The Impact of Gold Price on Bitcoin Price: Evidence from an ARDL Approach." SSRN Working Paper.

BAHMANI-OSKOOEE, M. AND A. GELAN. 2020. "Asymmetric Cointegration and the J-Curve: Evidence from Commodity Trade between the US and Ethiopia." *Economic Analysis and Policy*, 68: 152–162.

BANERJEE, A., J.J. DOLADO, AND R. MESTRE. 1998. "Error-Correction Mechanism Tests for Cointegration in a Single-Equation Framework." *Journal of Time Series Analysis*, 19(3): 267–283.

BOUOIYOUR, J. AND R. SELMI. 2015. "What Does Bitcoin Look Like?" *Annals of Economics and Finance*, 16(2): 449–492.

CHEVALLIER, J. 2017. "Bitcoin Price and Volatility: An Overview." *International Journal of Business*, 22(3): 173–191.

CHO, J.S., M. GREENWOOD-NIMMO, AND Y. SHIN. 2015. "Recent Developments in Nonlinear Cointegration with an Application to the Purchasing Power Parity Hypothesis." *Journal of Econometrics*, 188(2): 462–474.

CIAIAN, P. AND D'A. KANCS. 2019. "Cryptocurrencies: A Systematic Review." *Journal of Economics and Business*, 106: 105838.

CIAIAN, P., M. RAJCANIOVA, AND D'A. KANCS. 2016. "The Economics of Bitcoin Price Formation." *Applied Economics*, 48(19): 1799–1815.

DYHRBERG, A.H. 2016. "Bitcoin, Gold and the Dollar — A GARCH Volatility Analysis." *Finance Research Letters*, 16: 85–92.

ENGLE, R.F. AND C.W.J. GRANGER. 1987. "Co-Integration and Error Correction: Representation, Estimation, and Testing." *Econometrica*, 55(2): 251–276.

FOUSEKIS, P., V. GRIGORIADIS, AND N. TRAKAS. 2021. "The Dynamics of Cryptocurrency Prices: A Nonlinear ARDL Approach with Structural Breaks." *Journal of Economic Asymmetries*, 24: e00215.

JOHANSEN, S. 1988. "Statistical Analysis of Cointegration Vectors." *Journal of Economic Dynamics and Control*, 12(2–3): 231–254.

KREMERS, J.J.M., N.R. ERICSSON, AND J.J. DOLADO. 1992. "The Power of Cointegration Tests." *Oxford Bulletin of Economics and Statistics*, 54(3): 325–348.

KRIPFGANZ, S. AND D.C. SCHNEIDER. 2020. "Response Surface Regressions for Critical Value Bounds and Approximate p-Values in Autoregressive Distributed Lag (ARDL) Models." *Stata Journal*, 20(4): 929–966.

KRISTOUFEK, L. 2013. "BitCoin Meets Google Trends and Wikipedia: Quantifying the Relationship Between Phenomena of the Internet Era." *Scientific Reports*, 3: 3415.

KYRIAZIS, N.A. 2019. "A Survey on Bitcoin Price Drivers." *International Journal of Economics and Business Research*, 18(4): 389–411.

LAHIANI, A. AND B.D. SIMO-KENGNE. 2021. "Asymmetric Effects of Investor Sentiment on Cryptocurrency Returns: Evidence from a Nonlinear ARDL Approach." *Finance Research Letters*, 41: 101846.

MCNOWN, R., W.K. WONG, AND C.Y. SAM. 2018. "A New Test for Cointegration with Both I(0) and I(1) Variables." *Journal of Applied Econometrics*, 33(4): 618–629.

PESARAN, M.H. AND Y. SHIN. 1998. "An Autoregressive Distributed-Lag Modelling Approach to Cointegration Analysis." *Econometric Society Monographs*, 31: 371–413.

PESARAN, M.H., Y. SHIN, AND R.J. SMITH. 2001. "Bounds Testing Approaches to the Analysis of Level Relationships." *Journal of Applied Econometrics*, 16(3): 289–326.

PHILIPS, A.Q. 2018. "Have Your Cake and Eat It Too? Cointegration and Dynamic Inference from Autoregressive Distributed Lag Models." *American Journal of Political Science*, 62(1): 230–244.

PHILLIPS, P.C.B. AND B.E. HANSEN. 1990. "Statistical Inference in Instrumental Variables Regression with I(1) Processes." *Review of Economic Studies*, 57(1): 99–125.

REHMAN, M.Z.U. AND X.V. VO. 2021. "Asymmetric Effects of Crude Oil on Bitcoin: An NARDL Analysis." *Resources Policy*, 72: 102063.

SHIN, Y., B. YU, AND M. GREENWOOD-NIMMO. 2014. "Modelling Asymmetric Cointegration and Dynamic Multipliers in a Nonlinear ARDL Framework." In *Festschrift in Honor of Peter Schmidt*, 281–314. Springer.

STOCK, J.H. AND M.W. WATSON. 1993. "A Simple Estimator of Cointegrating Vectors in Higher Order Integrated Systems." *Econometrica*, 61(4): 783–820.

A Appendix: Additional Robustness Results

A.1 Sub-Sample Long-Run Coefficients

Table 3 presents the full long-run coefficient estimates for the pre-2020 and post-2020 sub-samples. The consistency of asymmetry direction across sub-samples supports the stability of our main findings.

Table 3: Sub-Sample Long-Run Coefficients

	Pre-2020 ($N = 1{,}973$)		Post-2020 ($N = 2{,}172$)	
Variable	L^+	L^-	L^+	L^-
ln_hash_rate	1.46	−0.51	0.41	0.24
ln_google_trends	1.13	0.67	0.18	0.11
ln_transactions	−1.05	0.17	0.94	0.94
ln_velocity	−0.26	−0.44	1.10	1.09
ln_gold_price	−4.48	−2.35	−2.99	−3.17
ln_cpi	−57.61	−133.83	−52.80	−700.11

Notes: Both sub-samples estimated using the NARDL(4,4) specification with structural break dummies. Pre-2020: January 2015–December 2019. Post-2020: January 2020–May 2026.

A.2 Specification Details

The baseline NARDL(4,4) model includes 73 regressors: a constant, one lagged level of the dependent variable (y_{t-1}), 12 lagged partial-sum levels (6 variables $\times$ 2 sign components), 4 lags of Δy, 48 lagged differences of the decomposed regressors (4 lags $\times$ 12 components), 7 structural break dummies, and a linear trend. The effective sample after lag construction is 4,150 daily observations.